

[2) Give an explicit formula for a function from the set of integers to the set of positive integers that is

a) one-to-one, but not onto.

Solution: $f(n) = \begin{cases} n^2 & \text{when } n \leq 0 \\ n^2 + 1 & \text{when } n \geq 1 \end{cases}$

We know that $\forall n \in \mathbb{Z}$, $n^2, n^2 + 1 \in \mathbb{Z}^+$. Here when $n = 0$, $n^2 + 1 = 1 \in \mathbb{Z}^+$

[Let $x, y \in \mathbb{Z}$ s.t. $f(x) = f(y)$

[We know that either both have the same side of the number line or ~~case~~ ^{they} belongs to the opposite side.

Case 1: $x < 0$ and $y < 0$. $\therefore x^2 = y^2 \Rightarrow (x-y)(x+y) = 0$
But, $x+y < 0$. $\therefore x-y = 0 \Rightarrow x = y$

Case 2: $x \geq 0$ and $y < 0$. $\therefore x^2 + 1 = y^2 \Rightarrow y^2 - x^2 = 1$

(Wrong attempt)